DAY-3

NAT and PAT ConceptsTheory :What is NAT/PAT? Importance in IP address conservation.Practical :Configure NAT and PAT on a router.Task: Simulate external connectivity using NAT/PAT.

[Q-1] In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.

(A)

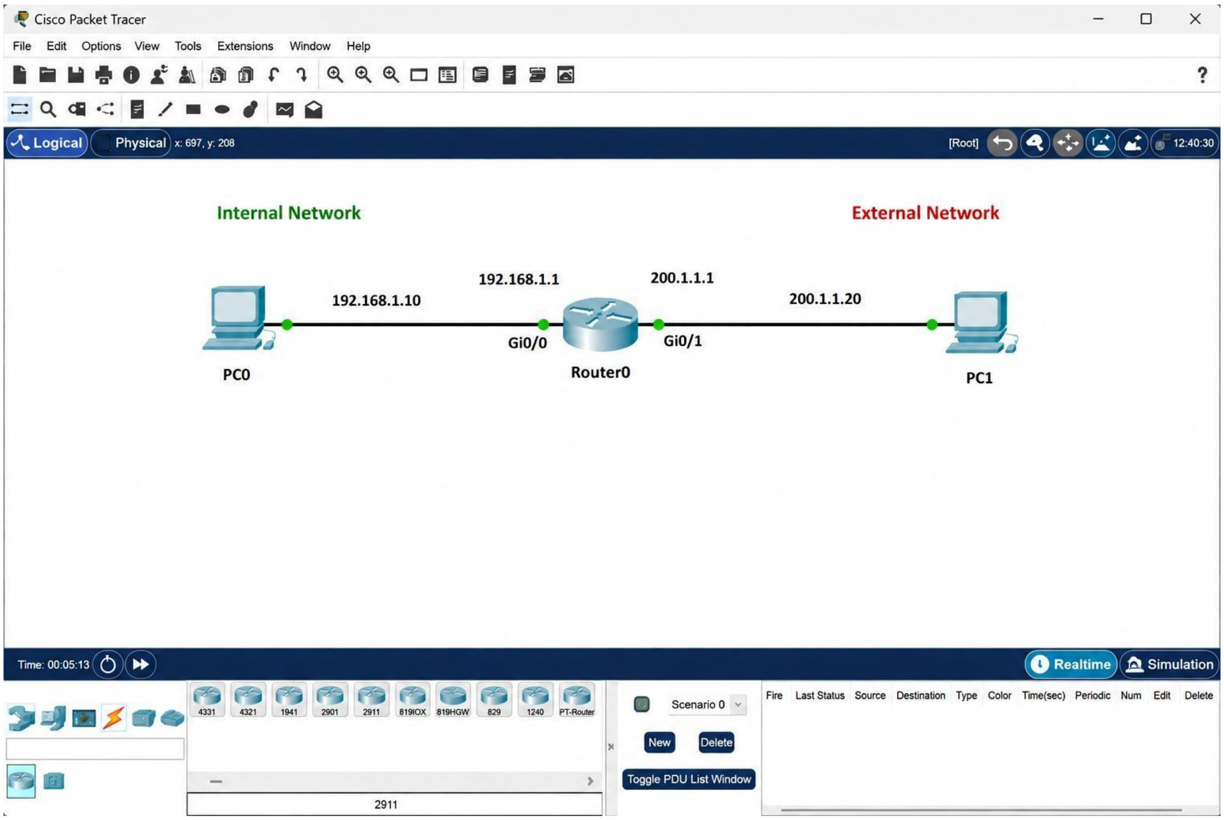
- NAT is a technology that translates one IP address into another. It is commonly used to translate **private IP addresses** inside a local network into **public IP addresses** for communication with external networks.

- PAT is a type of NAT that allows **multiple private devices** to share **one public IP address**. It distinguishes connections using different port numbers. PAT is also called **NAT Overload.**

- **Importance of NAT/PAT**
  - Conserves limited IPv4 public addresses.
  - Allows private networks to access the Internet.
  - Hides internal IP addressing from external networks.
  - PAT allows many users to share a single public IP address.

- **Steps :**

Step 1: Configure Internal PC
Step 2: Configure External PC
Step 3: Configure Router Interfaces
Step 4: Configure Static NAT



[Q-2] Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address. Hint: Use the 'ip nat inside source list' command with 'overload' to enable PAT.

[A]

- **PAT Router Configuration**

```
enable
configure terminal

interface gigabitEthernet0/0 ip
address 192.168.1.1 255.255.255.0 ip
nat inside
no shutdown
exit
```

interface gigabitEthernet0/1 ip
address 200.1.1.1 255.255.255.0
ip nat outside no
shutdown exit

Create an access control list that identifies the internal network:
access-list 1 permit 192.168.1.0 0.0.0.255

- **Enable PAT using the outside interface address:**
  ip nat inside source list 1 interface gigabitEthernet0/1 overload

- **Save:** end
  write memory

➢ **PAT Topology**

```
                  PUBLIC NETWORK
                         |
                     Web Server
                     200.1.1.20
                         |
                      Router0
                 Outside: 200.1.1.1
                         |
                -------------------------
|                         |
    PC0: 192.168.1.10       PC1: 192.168.1.11
                Internal Private Network
```

[Q-3] Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.

(A)

- **Test Connectivity**

Ping the external PC:
ping 200.1.1.20

If the NAT configuration is correct, the ping should be successful.
Example:
Reply from 200.1.1.20: bytes=32 time<1ms TTL=128
Reply from 200.1.1.20: bytes=32 time<1ms TTL=128
Reply from 200.1.1.20: bytes=32 time<1ms TTL=128
Reply from 200.1.1.20: bytes=32 time<1ms TTL=128

Check NAT Translation Table

On Router0:
show ip nat translations

For static NAT, an example result is:

Pro	Inside global	Inside local	Outside local	Outside global
---	200.1.1.10	192.168.1.10	---	---

This proves:
Inside Local: 192.168.1.10
↓ NAT
Inside Global: 200.1.1.10

You can also check NAT statistics:
show ip nat statistics

[Q-4] Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.

(A).

- **Add a Third Internal PC**
  - Add **PC2** to the internal network.
  - Because all three PCs need to connect to the router, you can add a **switch** between the PCs and the router.
- **Updated Topology**

```
PC0 ── 192.168.1.10 ──┐
          |
PC1 ── 192.168.1.11 ──┼── Switch ── Router0 ── External Server
          |           |
PC2 ── 192.168.1.12 ──┘     Public IP: 200.1.1.1
```

-

Complete PAT Configuration

```
enable
configure terminal

! Configure internal LAN interface
interface gigabitEthernet0/0 ip
address 192.168.1.1 255.255.255.0 ip
nat inside
no shutdown
exit

! Configure external/public interface
interface gigabitEthernet0/1 ip
address 200.1.1.1 255.255.255.0
ip nat outside
no shutdown
exit

! Allow all internal devices for NAT
access-list 1 permit 192.168.1.0 0.0.0.255

! Enable PAT/NAT Overload
ip nat inside source list 1 interface gigabitEthernet0/1 overload

end write
    memory
```

[Q-5] Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

(A)

- NAT and PAT are important because millions of users and devices need to communicate over networks, while public IPv4 addresses are limited.

- For example, imagine an office where 500 employees are using the Internet. Without PAT, the company could potentially require many public IP addresses. With PAT, all 500 users can access websites using a much smaller number of public IP addresses, or even one public IP in some network designs. The router uses port numbers to identify which internal user's traffic belongs to which connection.

- Similarly, applications and large companies use private addressing inside parts of their networks and carefully managed public addressing at network boundaries. NAT helps translate between these addressing spaces.